Human Factors in Interactive Dining

Designing an Interactive Dining Table for Modern Cafes

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Course: Human Factors in Interaction Design | Guided By: Prof. Umang Shah





Topic Introduction

Cafes today are not only places to eat but also social spaces where people interact, work, and spend time together.

With the increasing integration of technology in dining environments, interactive digital tables are emerging as a new way to enhance ordering, engagement, and user experience.

This project explores how human factors such as ergonomics, cognitive load, and social behavior influence the design of an interactive dining table for modern cafes.

Current State of Interactive Dining in India

- Interactive dining technologies are emerging to enhance restaurant experiences.
- Some global restaurants use **projection-based interactive tables**, such as **Le Petit Chef**, where animations appear directly on the dining table.
- **Fully interactive dining tables are still rare**, especially in India.
- In most Indian cafés, the primary digital interaction is **QR-code based menus**, where customers scan codes to view and order food





Secondary Research

Understanding human factors in café dining through research on ergonomics, cognitive behaviour, and digital dining technologies.

Physical Ergonomics

- Comfortable reach zone for diners: **350–400 mm**
- Average elbow space needed: **600–650 mm**
- Reaching the center of large tables can be uncomfortable

Cognitive Ergonomics

- Large menus increase **decision fatigue**
- Users struggle to **compare multiple dishes**
- **Visual cues** help faster decision making

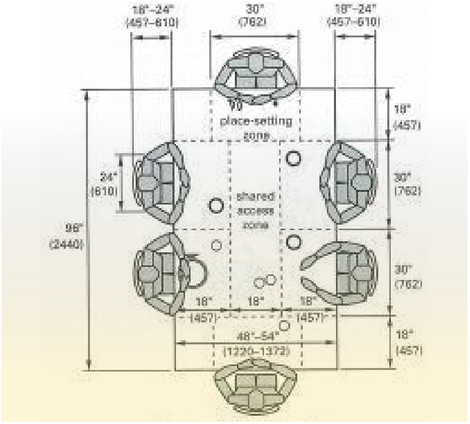
Social Dining Behaviour

- Dining in cafés is often a **group activity**
- People usually **discuss** before ordering
- **Food sharing** is very common

Digital & Interactive Dining Experiences

- **QR menus, touchscreen tables, projection mapping, AR, and sensor-based interfaces** enable interactive dining experiences.
- Cafés increasingly use these technologies to **enhance engagement, shared activities, and cultural experiences.**

Secondary Research Breakdown



Human Reach Zones

Anthropometric studies showing comfortable reach zones and elbow space in dining environments.



Menu Complexity

Research on cognitive load and decision fatigue in menu design.



Communal Dining Behaviour

Studies highlighting collaborative ordering and food sharing patterns.



Digital Ordering Systems

Examples of QR menus and touchscreen ordering systems in modern cafés.

Primary Research

Survey responses and café interviews were conducted to understand user behaviour and human factors influencing café dining experiences.

Social Dining Behaviour

- Most users visit cafes primarily for: casual hangouts, meetings & social interaction
- Ordering is often a **collaborative activity**, with groups discussing options and sharing dishes.

Menu Decision Challenges

- Users experience difficulty choosing from large menus.
- Factors influencing their choices: food images, dish names & highlighted items
- This indicates the importance of **visual cues and structured menus.**

Interaction with the Table

- Some users mentioned:
  - leaning forward to reach items in the center
  - asking others to pass items
- This suggests that **table interaction zones must consider physical reach and comfort.**

Digital Dining Awareness

- Many users are already familiar with:
  - QR-code menus
  - digital ordering systems
- However, some users worry that **technology might distract from social interaction.**

Primary Study Breakdown



Observation 1

The table surface quickly becomes occupied by multiple objects such as drinks, trays, laptops, and personal items.



Observation 2

Users place personal items like phones and laptops directly in front of them while interacting with their devices.



Observation 3

When the table becomes crowded, users use nearby surfaces such as chairs to place personal items.



Observation 4

Users engage in discussions and conversations before and during ordering.

User Interviews – Insights from Cafe Visitors



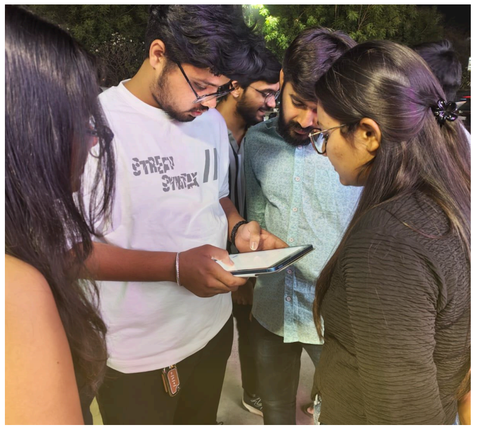
Insight 1

Most participants visit cafés primarily for **hangouts, dates, or family outings** rather than only for food.



Insight 2

During discussions, people naturally **show content on their phones or tablets to others around them.**



Insight 3

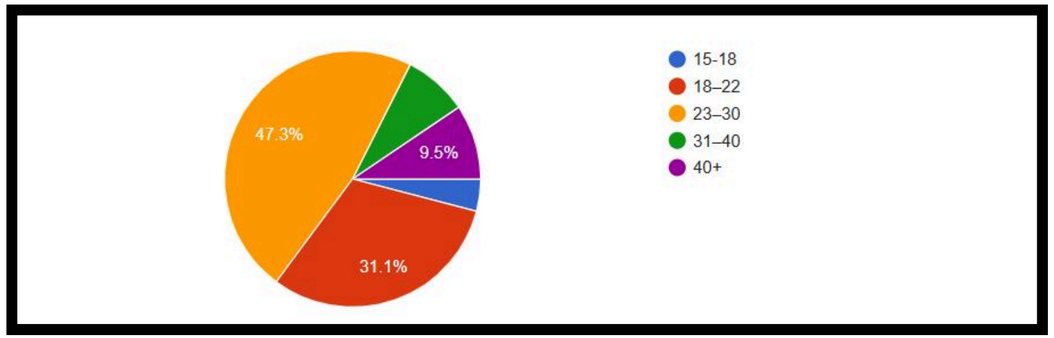
Participants mentioned **asking others to pass items or leaning forward to reach objects.**



Insight 4

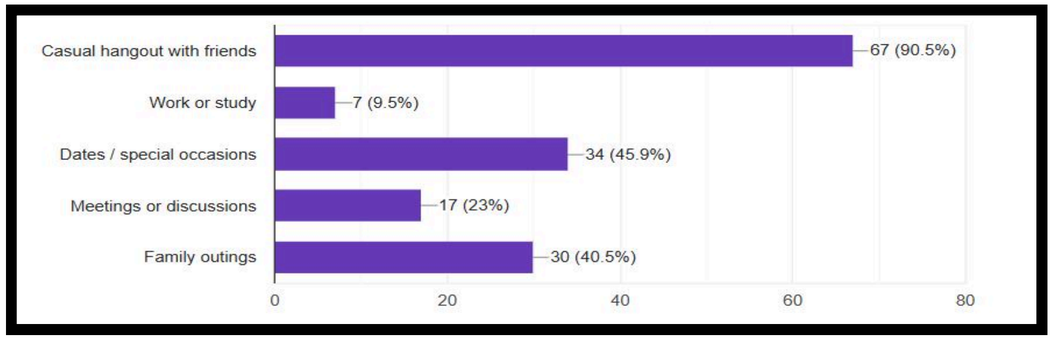
Many participants reported being **comfortable with digital interfaces**, but they still prioritize conversation with others.

Primary Research: Survey Insights



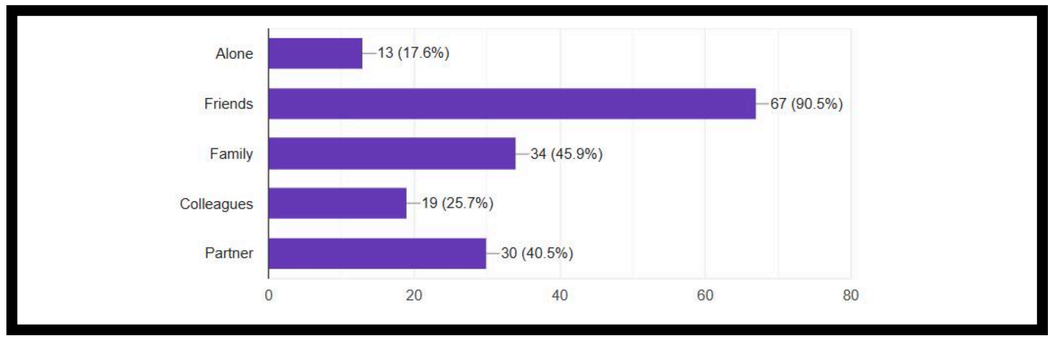
Age Group

Most cafe visitors are young adults (18–30).



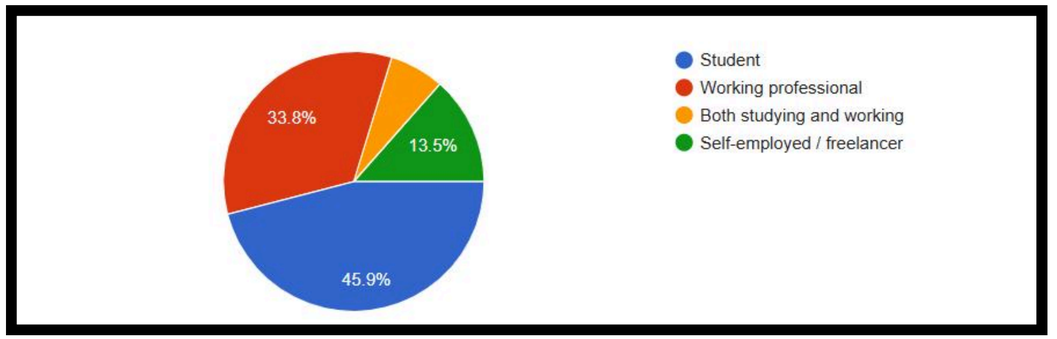
Occasion for Visiting Cafes

Cafes are mainly visited for casual hangouts with friends.



Who People Visit Cafes With

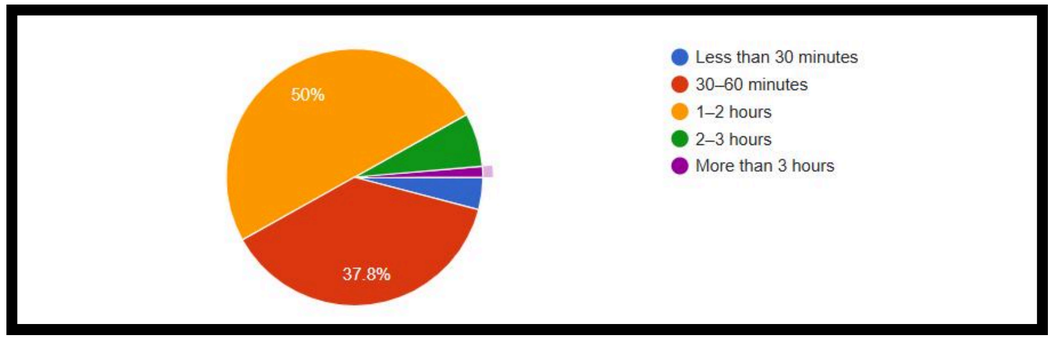
Most users visit cafes with friends, family & partners.



Occupation

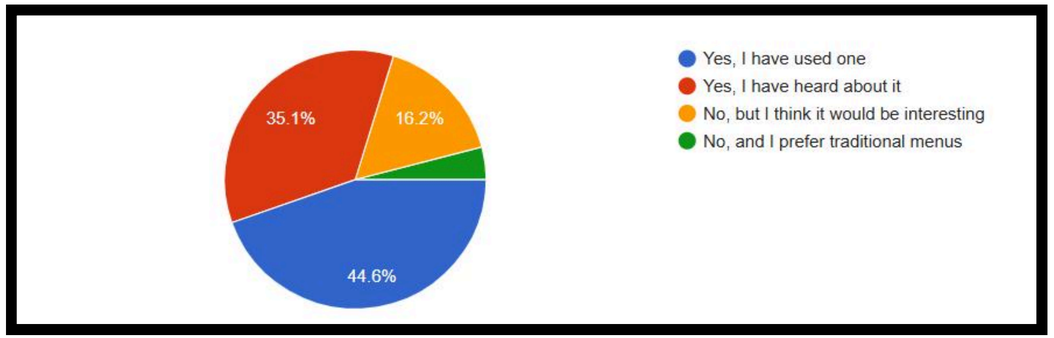
Most cafe visitors are students or young professionals.

Primary Research: Survey Insights



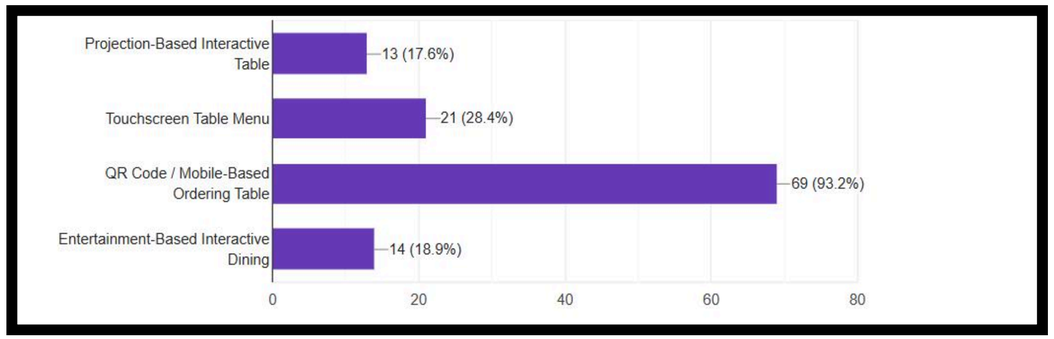
Time Spent in Cafe

Most users stay at cafes for 1–2 hours.



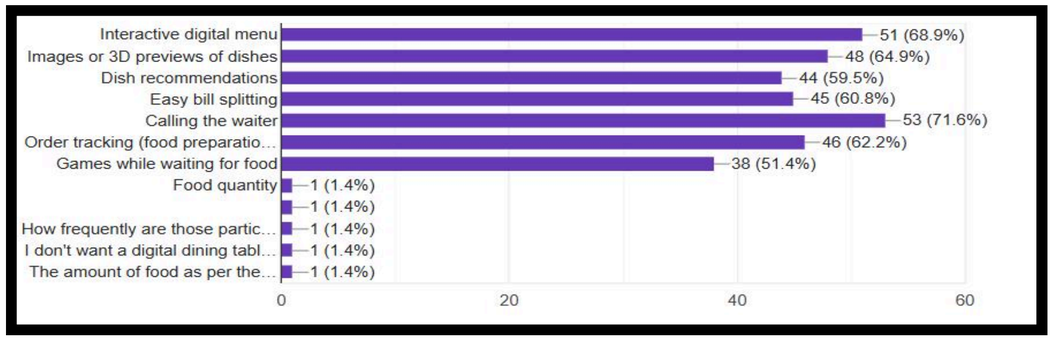
Awareness of Interactive Dining Tables

Many users are familiar with digital dining technologies.



Type of Interactive Dining Technology Known

QR code–based ordering is the most recognized technology.



Preferred Features for Interactive Tables

Users showed strong interest in features that improve convenience and interaction

Key User Groups Identified

Social Café Visitors (Students)

Age: 18–25

Profile:
- Visit cafes with friends
- Stay longer (1–2 hours)
- Share food and discuss orders

Needs:
- Fun dining experience
- Easy group ordering
- Interactive engagement

Young Professionals

Age: 23–35

Profile:
- Visit cafes for meetings or quick hangouts
- Limited time
- Prefer efficient service

Needs:
- Quick ordering
- Clear menu navigation
- Minimal waiting time

Family / Casual Visitors

Age: 30+

Profile:
- Visit occasionally with family
- Prefer simple dining experience

Needs:
- Clear menu information
- Comfortable interaction







Riya Sharma

Age: 22 Occupation: Student Location: Ahmedabad

About

Riya is a college student who frequently visits cafes with friends to relax, socialize, and try new food. She enjoys exploring different menu items and often shares dishes with her group.

Goals

- Enjoy a social dining experience
- Explore different food options
- Order quickly without confusion

Behavior

- Visits cafes with friends
- Discusses menu options before ordering
- Shares dishes with the group

Pain Points

- Large menus are confusing
- Ordering sometimes takes too long
- Reaching items in the center of the table can be inconvenient

Needs

- Clear and simple menu navigation
- Easy group ordering
- Comfortable table interaction

Empathy Map — Cafe Dining Experience

Says

- “What should we order?”
- “Let’s share something.”
- “What’s good here?”
- “This looks nice.”

Thinks

- Wants to try different dishes
- Looks for popular or recommended items
- Prefers simple and quick ordering
- Wants the experience to remain social and enjoyable

Does

- Discusses menu options with friends
- Shares food across the table
- Uses phones or QR codes to view menus
- Leans forward or asks others to pass items

Feels

- Social and relaxed when dining with friends
- Slightly overwhelmed by large menus
- Curious about new dishes
- Occasionally frustrated when ordering takes too long

How Might We Statements

Based on insights from secondary research, primary research, and user observations, the following design opportunities were identified to improve the café dining experience.

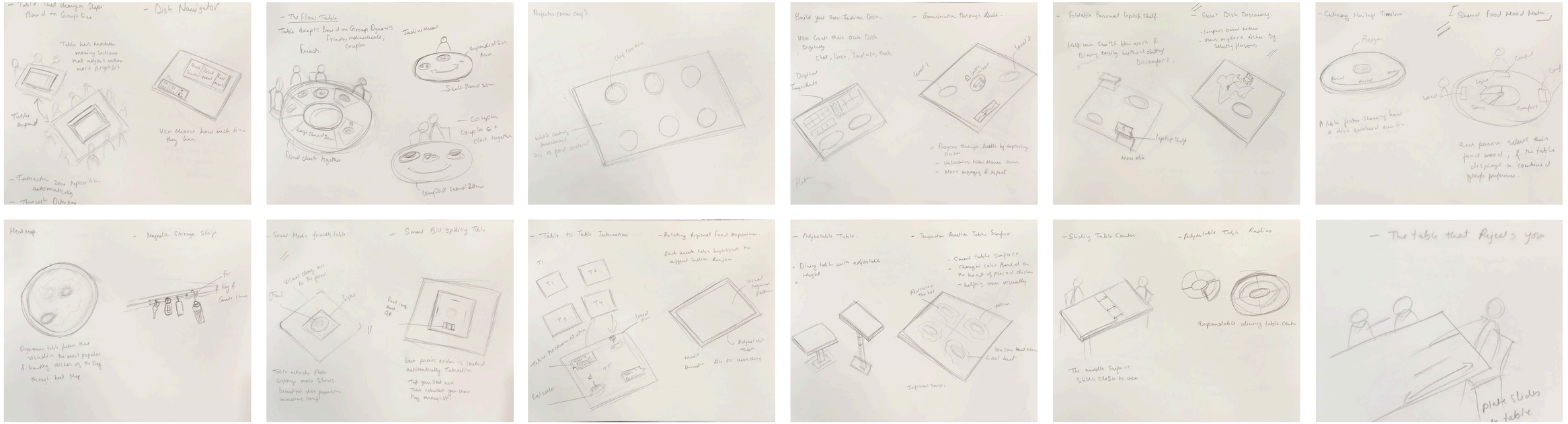
“How might we design an interface dining experience for genz/working professionals that supports social media buzz, quick decision making for ordering a smooth, socially enriching experience in modern Indian cafe.”

Design an **interactive dining table experience** for **modern Indian cafes** that enhances **group dining** by supporting **collaborative ordering, quick decision-making, and engaging social interaction.**

The system should consider **human factors** such as **physical ergonomics, cognitive load, and social behavior** to create a **seamless and comfortable dining experience.** It should encourage **participation, simplify choices**, and enable **natural, shareable moments** while preserving the **social and cultural essence** of café dining.

Design Brief

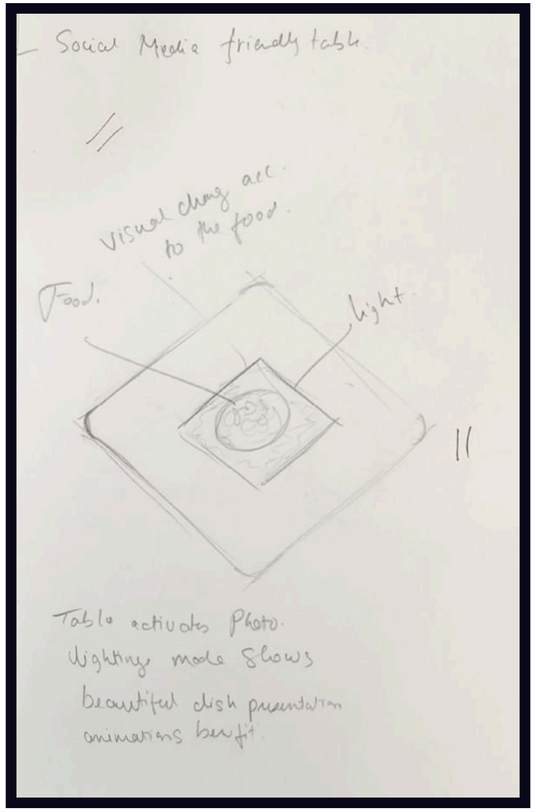
Ideation Sketches



Concept Shortlisting

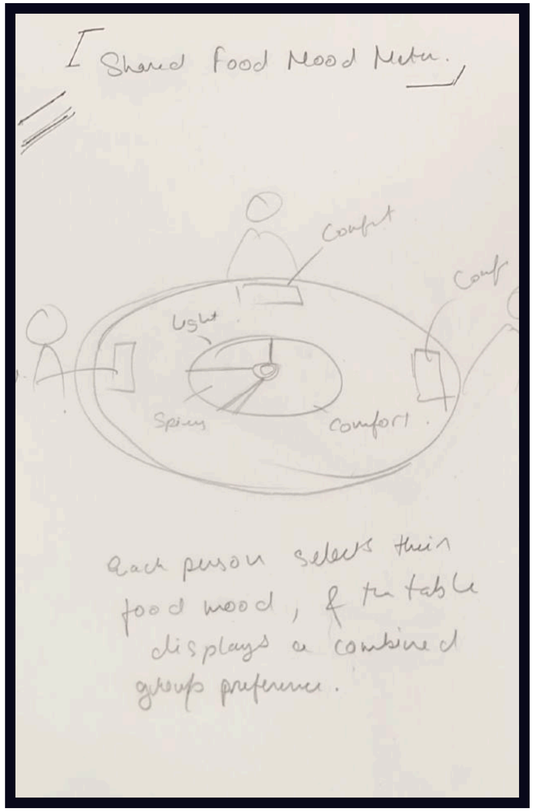
From the initial ideation phase, concepts were evaluated on the basis of feasibility, user engagement, and alignment with the HMW. Three key directions were shortlisted for further development and refinement.

Shortlisted Concept Directions



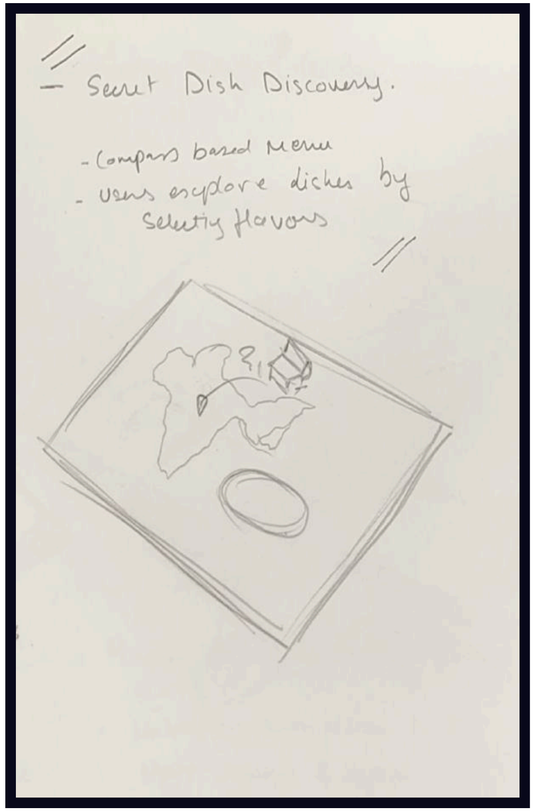
Social Media Friendly Table

Enhances shareable, aesthetic dining moments



Shared Food Mood Meter

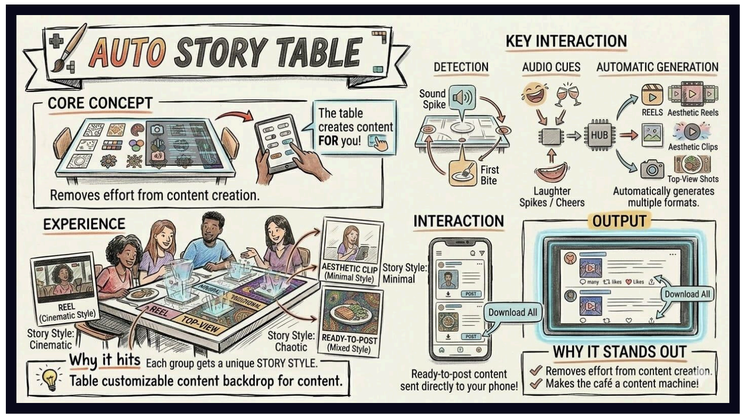
Encourages exploration through hidden menu interactions

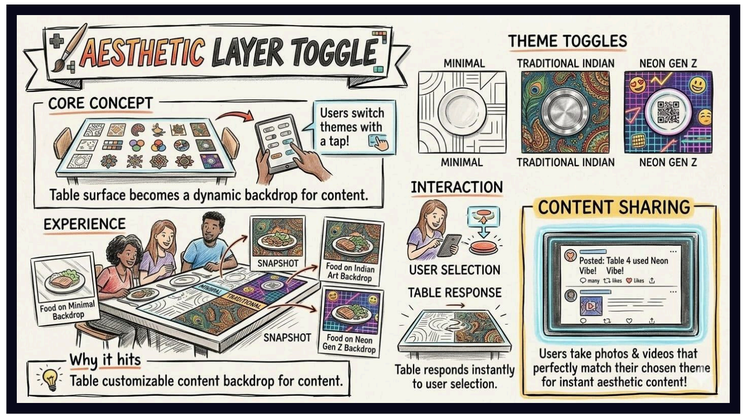


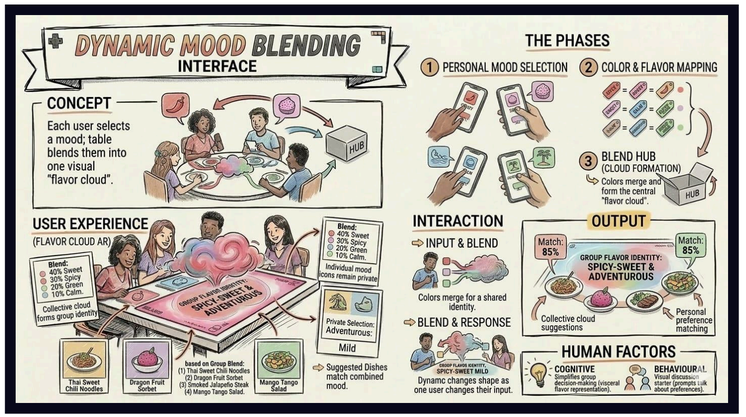
Secret Dish Discovery

Combines individual moods into a shared food direction

Final Concept Directions







Auto Story Table

Enhances shareable, aesthetic dining moments

- Automatically captures key interactions (first bite, cheers, reactions)
- Generates ready-to-post content
- Reduces effort in content creation

Aesthetic Layer Toggle

Transforms the table into a customizable visual experience

- Switch between themes (minimal, traditional, neon)
- Adapts to user mood and setting
- Creates a dynamic dining atmosphere

Dynamic Mood Blending

Combines individual preferences into a shared group decision

- Visual “flavor cloud” representing group mood
- Simplifies ordering process
- Encourages collaborative interaction



Auto Story Table

Concept

An interactive dining table that automatically captures and curates key dining moments, allowing users to stay present while creating shareable memories.

Interaction

- Detects key moments (first bite, cheers, laughter)
- Captures top-view and candid shots
- Generates short reels and aesthetic clips
- Sends ready-to-share content to users’ phones

User Value

- Reduces effort of content creation
- Encourages users to stay off their phones
- Creates memorable and shareable experiences

Human Factors

- **Cognitive**: Reduces mental load of capturing moments
- **Behavioral**: Encourages real-time social interaction
- **Physical**: Hands-free interaction, no device handling required



Dynamic Mood Blending Interface

Concept

A system where each user selects their food mood, and the table blends them into a shared visual “flavor cloud” to guide group decision-making.

Interaction

- Users select moods (spicy, sweet, light, comfort, etc.)
- Colors merge into a shared visual output
- Table suggests dishes based on the combined mood

User Value

- Simplifies group decision-making
- Reduces confusion and ordering time
- Encourages collaborative participation

Human Factors

- **Cognitive:** Reduces decision fatigue
- **Behavioral:** Promotes group discussion and consensus
- **Physical:** Simple, intuitive input within reach

Aesthetic Layer Toggle



Concept
A customizable table interface that allows users to switch between different visual themes, enhancing the dining ambience and experience.

Interaction
- Users toggle between themes (minimal, traditional, neon, etc.)
- Table surface adapts visually in real-time
- Enhances food presentation and table aesthetics

User Value
- Personalizes dining experience
- Enhances visual appeal for social sharing
- Adapts to mood and occasion

Human Factors
- **Cognitive:** Simple and intuitive customization
- **Behavioral:** Encourages engagement and exploration
- **Emotional:** Creates a more immersive dining atmosphere



60–70% leaned towards **Auto Story Table**

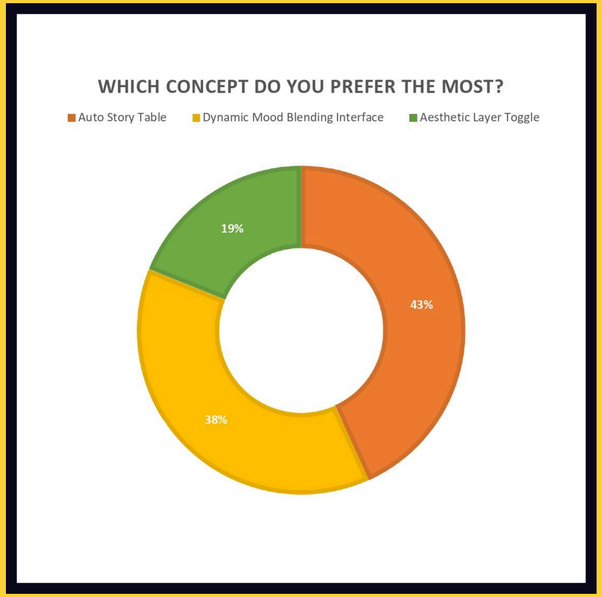
User Validation Insights

Majority users preferred **Auto Story Table** for capturing memories effortlessly

Dynamic Mood Blending was valued for **faster decision-making**

Aesthetic Layer Toggle enhanced **ambience and visual experience**

Users want **less phone usage during dining**



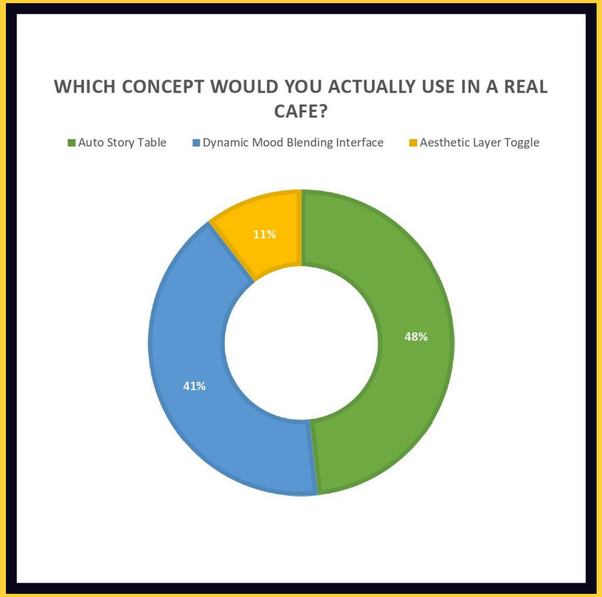
Key Learnings from User Feedback

What Worked:

- Users value **effortless memory capture**
- Strong interest in **social + shared experiences**
- Need for **quick and simple decision-making**

Concerns:

- **Privacy issues** with automatic recording
- Users want **control over what is captured**
- Over-automation can feel intrusive





Auto Story Table (Refined)

Key Refinements Based on Feedback

- Added user **control over content capture** to address privacy concerns
- Introduced **opt-in / opt-out** interaction for comfort and trust
- Reduced automation to avoid intrusive experience
- Focused on **subtle, non-disruptive interaction cues**

Improved Experience

- More user-controlled and transparent system
- Balances automation with comfort
- Enhances social interaction without intrusion

Design Impact

- Builds trust and acceptance among users
- Maintains engagement while respecting privacy
- Creates a more realistic and implementable solution

Figma Link

https://relate-gong-21547507.figma.site

Human Factors & System Integration

Sensing System (Camera & Gimbal)

- Overhead camera ensures **full table visibility without obstruction**
- Gimbal enables **smooth, non-intrusive motion tracking**
- Positioned to avoid **blocking user space or eye contact**
- Designed for **subtle presence to reduce user discomfort**

Table Interaction & Physical Ergonomics

- All interactions within **comfortable reach zone (350–400 mm)**
- Table height optimized for **posture and dining comfort (700–750 mm)**
- Hands-free experience reduces **physical effort and fatigue**
- No interference with **food placement or personal space**

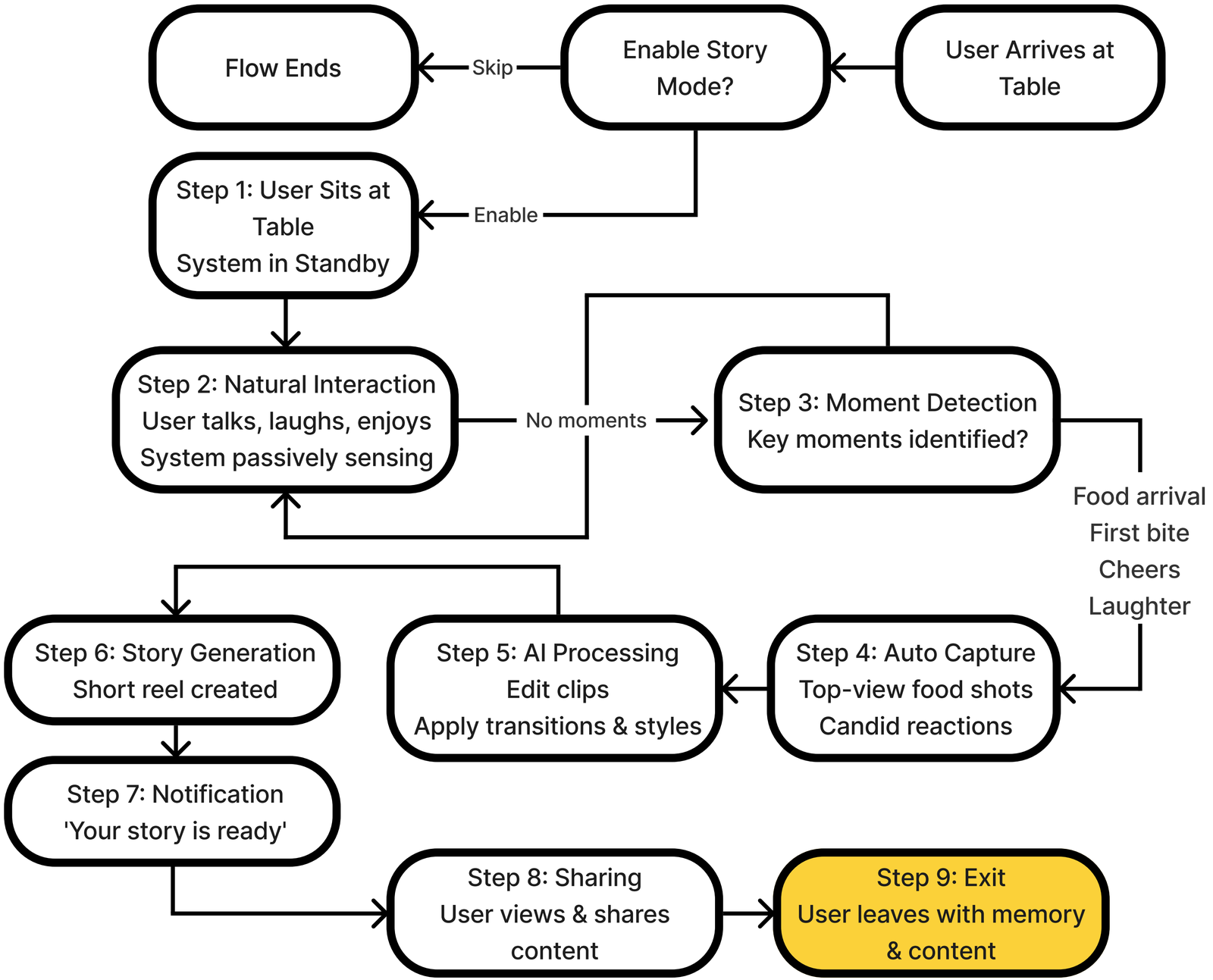
Tablet Interface (User Control System)

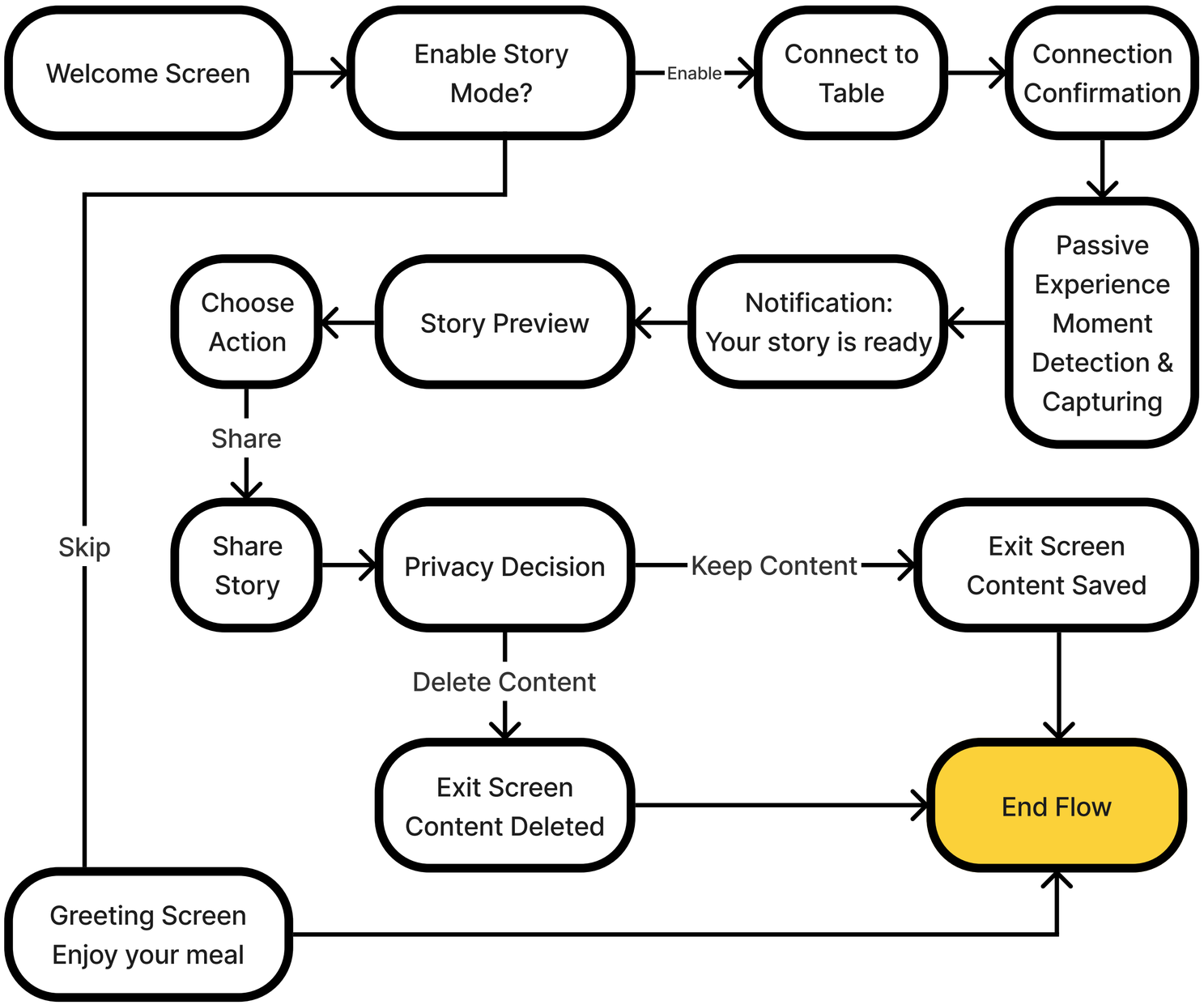
- Provides **clear control over content capture (opt-in/out)**
- Positioned at **easy viewing angle (30°–45°)**
- Simple, minimal UI for **low cognitive load**
- Enables **quick decisions without interrupting conversation**





Experience Flow





Information Architecture : In Table App

Next Steps & Future Scope

Advanced User Testing & Iteration

- Test across different cafe environments and group types
- Measure comfort, engagement, and usability
- Refine interaction based on real-world behavior

AI Personalization Enhancement

- Adapt story styles based on user preferences
- Improve emotion and moment detection accuracy
- Enable more context-aware content generation

Multi-Platform Integration

- Direct sharing to social media platforms
- Seamless phone connectivity and user control
- Integration with café systems for personalized experiences

Experience Expansion

- Extend to restaurants, lounges, and event dining
- Enable group-based interactive experiences across tables
- Build a larger “social dining ecosystem”

Designing for Human Behavior

- Learned to design beyond functionality, focusing on **user behavior and interaction patterns**
- Understood the importance of **social dynamics in group experiences**

Integrating Human Factors

- Applied **physical, cognitive, and behavioral ergonomics in design decisions**
- Balanced **comfort, usability, and engagement** within the dining context

Iterative Design Process

- Explored multiple concepts before converging on a final direction
- Used **user validation to refine and improve the concept meaningfully**

Key Learnings

Thank you!



Gauri Joshi & Riddhi Patel | Team In-Table